

● Vision

● Language

● Action

# Vision-Language- Action Models

“Pick up the red bottle and place it on the table.”

**8.4B**

PARAMETERS

**10Hz**

CONTROL RATE

**7-DOF**

ACTION SPACE

**~85ms**

INFERENCE

← → ARROW KEYS OR SPACE TO NAVIGATE

— THE PROBLEM

# Why Traditional Robots **Fail**

## ✗ TRADITIONAL ROBOTS

- ✗ Hard-coded, rigid behaviour
- ✗ Breaks when objects move
- ✗ Requires expert programming per task
- ✗ Cannot understand language

## ✓ VLA-POWERED ROBOT

- ✓ Understands natural language
- ✓ Adapts to scene changes in real-time
- ✓ Generalizes to new objects
- ✓ Learns from demonstrations

RGB Image + "Pick up the red bottle..." → 🦾 Robot Action

— CONCEPT

# What is a **VLA Model**?



## VISION

What does the robot **SEE**?

RGB camera captures the scene. Visual encoder extracts spatial features.

DINOv2-ViT-L/14



## LANGUAGE

What is the robot **TOLD**?

Natural language instruction. LLM understands meaning and intent.

LLaMA-3.1-8B



## ACTION

What should the robot **DO**?

7-DOF delta actions. Diffusion policy generates precise movements.

Diffusion Policy

VLA = **Foundation VLM** (pretrained on internet) + **Robot Action Head** (trained on demonstrations)

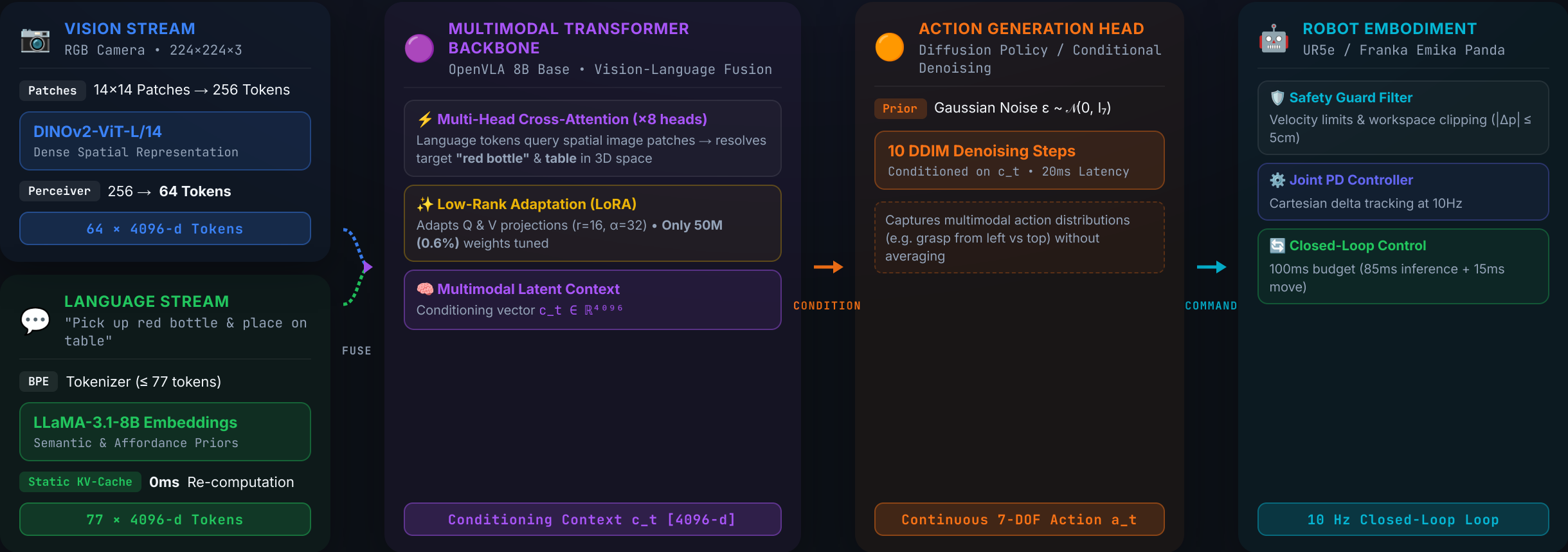
# Why I Chose OpenVLA

| MODEL                                  | PARAMS | ACTION HEAD   | OPEN?    | MY SELECTION      |
|----------------------------------------|--------|---------------|----------|-------------------|
| OpenVLA ← My Base                      | 7.5B   | Tokenized     | ✓ Open   | Backbone          |
| RT-2 (Google DeepMind)                 | 55B    | Tokenized     | ✗ Closed | Design inspo      |
| π <sub>0</sub> (Physical Intelligence) | 3B     | Flow-matching | ✗ Closed | Action head inspo |
| Octo                                   | 93M    | Diffusion     | ✓ Open   | Edge reference    |

✨ My Architecture: OpenVLA backbone + Diffusion Policy action head = Best of both worlds

— END-TO-END SYSTEM

# Proposed VLA Architecture



● Real-Time Closed-Loop Feedback: Robot executes  $a_t$  → physical state updates → camera captures  $I_{\{t+1\}}$  at 10Hz

Base: OpenVLA 8B • Trainable: 50M LoRA (0.6%) • Inference: ~85ms • Rate: 10 Hz

# Vision Encoder

## PROCESSING FLOW

```
RGB Image [224 × 224 × 3]
↓
14×14 patch split (16×16 grid)
↓
256 patch tokens × 1024-dim
↓
DINOv2-ViT-L/14
↓
Rich spatial features
↓
Perceiver Resampler
  256 → 64 tokens
↓
64 × 4096-dim
```

## WHY DINOv2 OVER CLIP?

CLIP → knows *WHAT* is in image

DINOv2 → knows *WHERE* objects are

**For grasping, spatial accuracy is everything.**

## PERCEIVER RESAMPLER

256 → 64 tokens → **4× faster** fusion

Learnable queries compress only task-relevant visual info.

From DeepMind's **Flamingo** (NeurIPS 2022).

# Language Encoder

## PROCESSING FLOW

```
"Pick up the red bottle  
and place it on the table"
```

↓

**BPE Tokenizer**

vocab = 32,000 tokens

↓

**Token IDs:**

```
[1, 29347, 701, 278, ...]
```

↓

**LLaMA-3.1-8B**

(8 billion parameters)

↓

**77 tokens × 4096-dim**

semantic embeddings

## WHAT LLAMA UNDERSTANDS

*"red"*

→ colour attribute

*"bottle"*

→ object class

*"pick up"*

→ action intent

*"on the table"*

→ spatial goal

## ⚡ KV-CACHE TRICK

Instruction is **static** across episode.

Compute once → cache → **reuse for 100+ steps**.

Saves **40ms per step** → critical for 10Hz.

# Multimodal Fusion

## CROSS-ATTENTION MECHANISM

```
Language: "red bottle"
  ↓ QUERIES →
Visual patches [64 tokens]
  ↓
High attention score
on bottle-region patches
  ↓
6× Self-Attention layers
  ↓
Global Avg Pool
  ↓
Scene Embedding [1 × 4096]
```

## VISUAL GROUNDING

Language words **search** through image patches.

*"red"* → high attention on red pixels

*"bottle"* → high attention on bottle shape

*"table"* → high attention on table surface

Words literally **find** their visual counterparts.

## ARCHITECTURE DETAIL

8-head cross-attention

6× transformer self-attention layers

Output: **1 × 4096 scene embedding**

= complete multimodal scene understanding

# Action Generation — Diffusion Policy

## WHY DIFFUSION? NOT REGRESSION?

Multiple valid grasps exist:

■ Grasp from left ■ Grasp from right

**MSE regression** → averages → bad middle grasp ❌

**Diffusion policy** → full distribution ✅



## 7-DOF ACTION SPACE

| DIM                                     | MEANING        | RANGE         |
|-----------------------------------------|----------------|---------------|
| $\Delta x, \Delta y, \Delta z$          | Position delta | $\pm 2$ cm    |
| $\Delta roll, \Delta pitch, \Delta yaw$ | Rotation delta | $\pm 5^\circ$ |
| gripper                                 | Open / close   | {0, 1}        |

## DDIM FAST SAMPLING

Training: 50 denoising steps

Inference: 10 DDIM steps

= 5× speedup, minimal quality loss

Time: ~20ms

— TRAINING

# 3-Stage Training Pipeline

## 1 FOUNDATION (FREE)

Pretrained on HuggingFace:

- **DINOv2**: 142M images
- **LLaMA-3.1-8B**: 15T tokens

Cost: \$0 — download weights

## 2 ROBOT CO-TRAINING

- Open X-Embodiment: **1M+ eps**
- BridgeData V2: **60K demos**
- DROID: **76K demos**

8× A100 GPUs, ~3 days

## 3 LORA FINE-TUNING

- **100–200** custom demos
- LoRA rank=16, alpha=32
- Only **0.6%** params trained

3–5 hours on 8× A100

Pretrained  
VLM weights



Robot data  
co-training



Task-specific  
LoRA tuning



 Deployed  
VLA Model

— EFFICIENCY

# LoRA — Fine-Tune **8B Params** in Hours

## ● Without LoRA

Parameters updated: **8,000,000,000**  
Training time: **Weeks**  
GPU memory: **Extreme**  
Risk: **Catastrophic forgetting**

## ● With LoRA (r=16) ✓

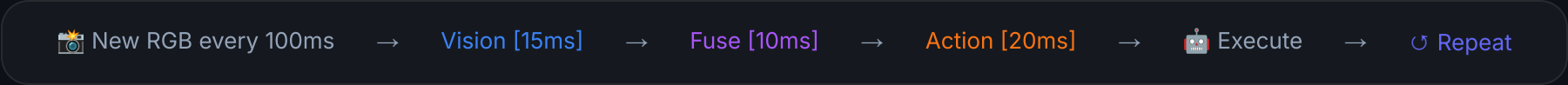
Parameters updated: **~50,000,000**  
Training time: **3–5 hours**  
GPU memory: **Manageable**  
Risk: **No forgetting**

$$W' = W + \Delta W = W + B \cdot A \quad \text{where } \text{rank}(A)=r=16 \ll d$$

Only A and B matrices are trained → **95% compute savings** → task-specific with 100–200 demos

— REAL-TIME

# Inference Pipeline — 10Hz Control



— DEPLOYMENT

# Real-World **Challenges** & Solutions

## **Latency (8B model too slow)**

TensorRT INT8 quantization + KV-cache + 10-step DDIM → 85ms total

## **Distribution Shift**

Domain randomization in simulation + real-world fine-tuning with 100-200 demos

## **Object Hallucination**

Parallel YOLO-v8 object detector + confidence thresholding → stop if uncertain

## **Safety**

Action clamping + velocity limits + force-torque monitoring → halt on anomaly

## **Edge Deployment**

Distill 8B → 3B + INT8 quantization + NVIDIA Jetson Orin (275 TOPS)

## **Sim-to-Real Gap**

MuJoCo/IsaacSim training + texture/lighting randomization + real demo mixing

— RESEARCH FOUNDATION

# Key Papers & References

|      |                                 |                                                |
|------|---------------------------------|------------------------------------------------|
| 2022 | Flamingo — DeepMind             | Cross-attention fusion, Perceiver Resampler    |
| 2022 | LoRA — ICLR                     | Efficient fine-tuning, low-rank adaptation     |
| 2023 | Diffusion Policy — RSS          | Action generation via denoising                |
| 2023 | RT-2 — Google DeepMind          | VLA design: tokenized actions + web data       |
| 2023 | DINOv2 — Meta                   | Vision encoder: spatial self-supervised ViT    |
| 2024 | OpenVLA — Open Source           | My selected backbone: 8.4B, DINOv2 + LLaMA-3.1 |
| 2024 | Open X-Embodiment               | 1M+ demos, 22 robots, 21 labs — training data  |
| 2024 | $\pi_0$ — Physical Intelligence | Flow-matching action head inspiration          |

— SUMMARY

# The Complete VLA Architecture

